Diurnal Temperature Range Trends in St. John's, NL: Testing for Monotonic Change Under Global Warming (2000–2025)

Isaac Adoboe *ieadoboe@mun.ca*
Department of Mathematics and Statistics
University of Newfoundland

Abstract

This study analyzes trends in diurnal temperature range (DTR) in St. John's, Newfoundland and Labrador from 2000 to 2025 to test whether the region exhibits declining DTR consistent with global warming theory. Daily temperature data from two meteorological stations were used to calculate DTR (defined as $T_{\max} - T_{\min}$) and assess trends using Mann-Kendall tests. Results show significant warming (+1.07°C over 26 years, $p < 0.001$) but no significant trend in annual mean DTR ($\tau = 0.107$, $p = 0.229$) or extreme DTR event frequencies. Seasonal analysis reveals weak DTR amplitude (3.6°C) with spring-fall asymmetry, indicating strong oceanic moderation. These findings suggest that maritime climates may exhibit delayed or dampened responses to global warming compared to continental regions, highlighting the importance of regional context in climate change assessment.

1 Introduction

1.1 Global Warming and Diurnal Temperature Range

Global mean surface temperature has increased by approximately 1.1°C since pre-industrial times, with warming rates accelerating in recent decades IPCC (2021). A key characteristic of this warming is its *asymmetric* nature: nighttime minimum temperatures ($T_{\min}$) are rising faster than daytime maximum temperatures ($T_{\max}$), leading to a global decline in diurnal temperature range (DTR) Karl et al. (1993). DTR, defined as:

$$\mathrm{DTR}(t) = T_{\max}(t) - T_{\min}(t) \tag{1}$$

serves as an important climate indicator because it reflects atmospheric conditions including cloud cover, humidity, and air mass characteristics. High DTR typically indicates clear skies and continental influence, while low DTR suggests cloudy, maritime conditions.

The physical mechanism behind DTR decline involves increased atmospheric moisture from warming, which enhances cloud cover and traps outgoing longwave radiation at night, thereby elevating $T_{\min}$ more than $T_{\max}$ Dai et al. (1999). This pattern has been observed globally, particularly in continental regions. However, maritime locations may exhibit different responses due to oceanic thermal inertia and persistent cloud cover.

1.2 Study Location and Motivation

St. John's, Newfoundland and Labrador (47.6°N, 52.7°W) provides an ideal test case for examining DTR trends in a maritime climate. The city is strongly influenced by the North Atlantic Ocean and the cold Labrador Current, resulting in moderate temperature extremes, year-round, high humidity & frequent cloud cover and significant oceanic moderation of diurnal temperature cycles.

This study addresses two key questions:

1. Does St. John's exhibit declining DTR consistent with global warming theory?
2. How do extreme DTR events (high and low) change over the study period?

Understanding regional climate responses is critical for adaptation and socioeconomic planning, as global patterns may not apply uniformly across different geographic contexts.

2 Data and Methods

2.1 Data Sources

Climate data were obtained from ClimateData.ca (Environment and Climate Change Canada) for the St. John's region. The dataset includes daily observations from two meteorological stations.

Station configuration:

- **Station 1 (St. John's A):** January 1, 2000 – March 19, 2012 (primary record)
- **Station 2 (St. John's West):** August 1, 2010 – November 30, 2025 (commissioned July 2010)

To avoid data duplication during the overlap period, only Station 1 data were used through to July 2010, after which Station 2 became the primary data source following Station 1's decommissioning. This sequential configuration maintains continuity while preventing artificial inflation of sample sizes during the transition. The combined dataset spanning January 1, 2000 to November 30, 2025 (26 years).

Key variables include:

- Daily minimum temperature ($T_{\min}$, °C)
- Daily maximum temperature ($T_{\max}$, °C)
- Mean temperature ($T_{\text{mean}} = (T_{\min} + T_{\max})/2$)
- and 33 other features including station descriptors, datetime information, meteorological features like total snow, total rain, wind gusts and flags for various meteorological features

After data preprocessing (removing 2 records where $T_{\min} > T_{\max}$, physically impossible), the final dataset contains 9,431 daily observations with less than 0.03% missing data.

2.2 DTR Calculation

DTR was calculated for each day as the difference between maximum and minimum temperatures. Regional DTR was computed by averaging across available stations. Summary statistics:

- Mean DTR: 7.85°C
- Standard deviation: 2.98°C
- Range: 0.20–19.80°C

2.3 Statistical Methods

2.3.1 Trend Detection

We employed the Mann-Kendall test, a non-parametric method for detecting monotonic trends in time series data. This test is robust to outliers and does not assume normally distributed residuals, making it ideal for climate data. The test statistic, Kendall's τ, ranges from -1 to $+1$, where:

- $\tau > 0$: Increasing trend
- $\tau < 0$: Decreasing trend
- $p < 0.05$: Statistically significant (rejecting null hypothesis of no trend)

Linear regression was also performed to quantify trend magnitude and provide confidence intervals, with the caveat that temporal autocorrelation may affect p-values.

2.3.2 Extreme Event Analysis

Extreme DTR days were defined using percentile-based thresholds computed from the entire dataset:

- **High-DTR extremes:** Days exceeding the 90th percentile (11.04°C)
- **Low-DTR extremes:** Days below the 10th percentile (5.18°C)

Annual frequencies of extreme events were calculated and tested for trends using Mann-Kendall tests. This approach tests whether the distribution tails are changing independently of mean shifts.

3 Results

3.1 Temperature Trends

Figure 1 demonstrates significant warming in St. John's over the study period. Mean annual temperature increased from approximately 4.5°C in 2000 to 5.6°C in 2025, representing a total change of 1.07°C. The linear regression yields a slope of 0.041 ± 0.009°C/year ($p < 0.001$), indicating that warming is statistically robust (significant) and explains 41.8% of inter-annual temperature variance.

This warming rate is consistent with regional climate projections for Atlantic Canada and confirms that St. John's is experiencing temperature increases comparable to global averages, despite its maritime setting.

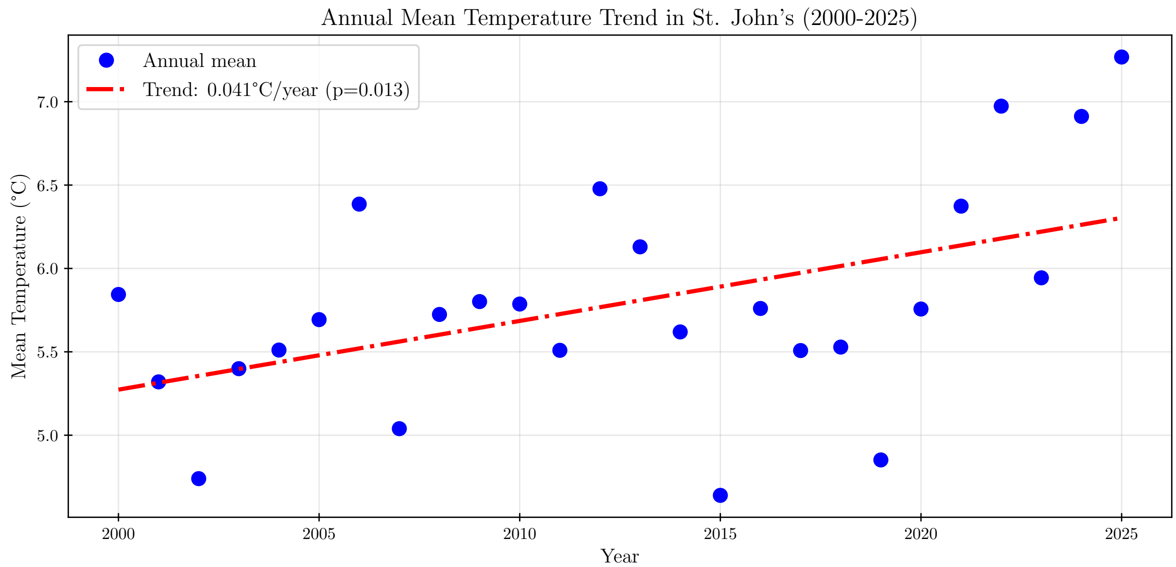

Figure 1: Annual mean temperature trend in St. John's (2000–2025). Blue circles represent annual mean temperatures, red dashed line shows linear regression fit. Significant warming of 1.07°C over 26 years is evident (slope = 0.041°C/year, $p < 0.001$, $R^2 = 0.418$).

3.2 Annual DTR Trends

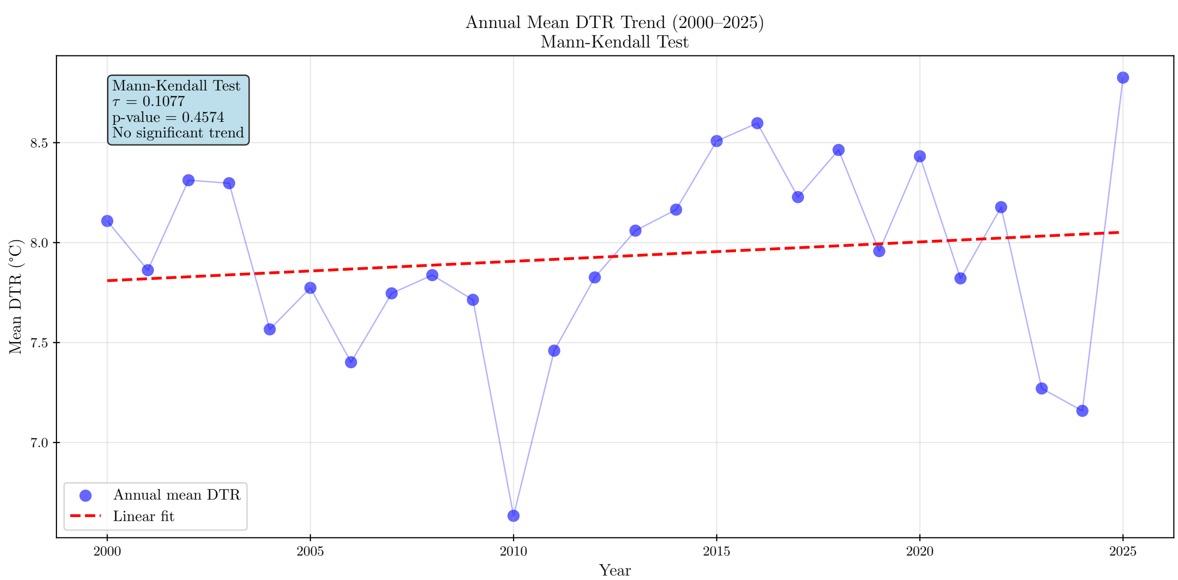

Figure 2: Annual mean DTR with Mann-Kendall test results (2000–2025). Blue circles show annual mean DTR, blue line connects consecutive years, red dashed line shows linear regression reference. Mann-Kendall test yields $\tau = 0.107$ ($p = 0.229$), indicating no significant monotonic trend despite visual scatter.

Despite significant warming, annual mean DTR shows no statistically significant trend (Figure 2). The Mann-Kendall test yields $\tau = 0.107$ with $p = 0.229$, failing to reject the null hypothesis of no monotonic trend at the $\alpha = 0.05$ level.

While Kendall's τ is slightly positive, suggesting a weak tendency toward increasing DTR, this is not statistically meaningful. High inter-annual variability (range: 6.5–9.5°C) dominates any underlying trend signal. This result is inconsistent with the global pattern of declining DTR under warming and suggests that maritime climates may respond differently to global forcing.

3.3 Seasonal DTR Patterns

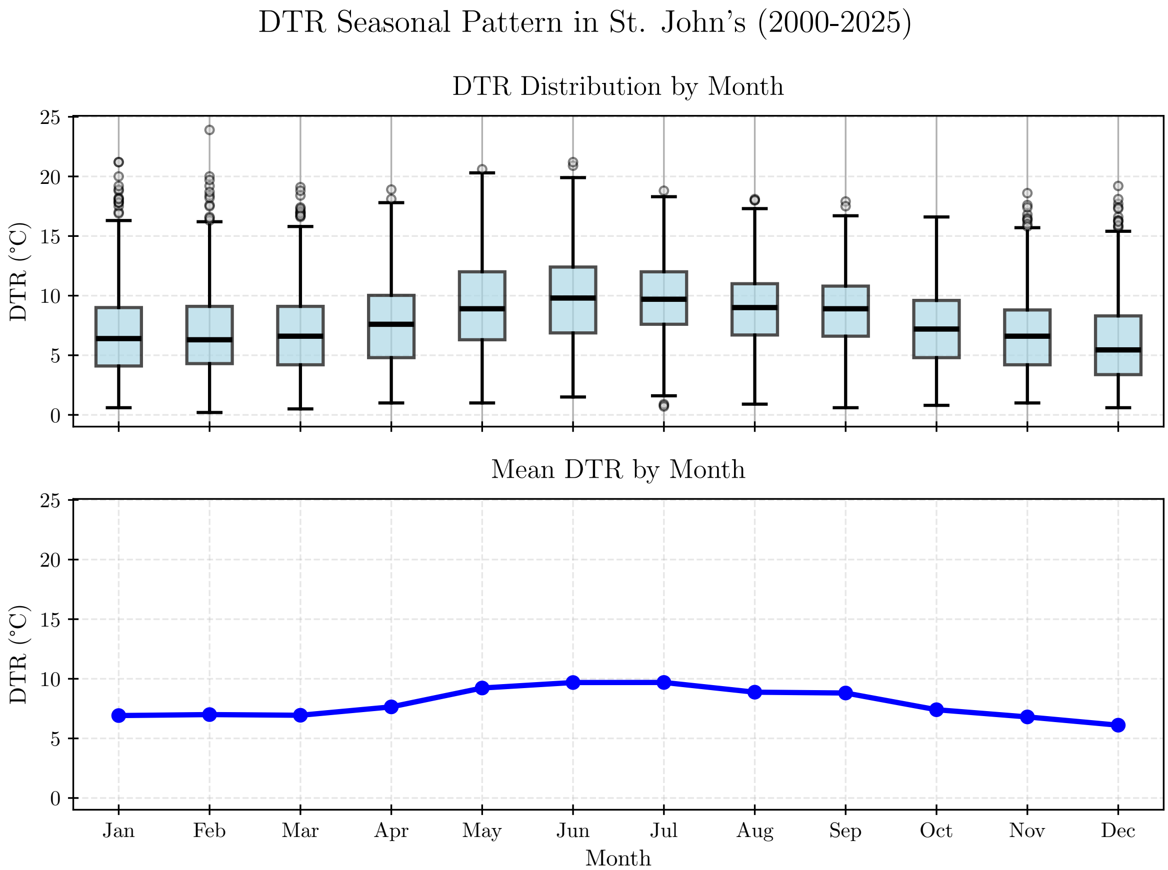

Figure 3: Seasonal DTR pattern in St. John's. Top panel: monthly DTR distribution (boxplot) showing median (dark blue line), interquartile range (boxes), and outliers. Bottom panel: mean DTR by month (blue line). Winter months exhibit suppressed DTR (6–7°C) due to maritime influence, while summer shows elevated DTR (9–10°C) reflecting continental air mass intrusion.

Figure 3 reveals a clear seasonal cycle in DTR with several notable features.

1. **Weak amplitude:** Seasonal DTR amplitude is only 3.6°C (range: 5.4–9.9°C), indicating strong year-round oceanic moderation. Continental sites typically exhibit amplitudes exceeding 10°C.

2. **Winter suppression:** December–February DTR averages 6.4°C, reflecting persistent cloud cover and maritime air masses that limit radiative cooling at night.

3. **Summer elevation:** June–August DTR peaks at 9.5°C as continental high-pressure systems bring clearer skies and stronger diurnal temperature cycles.

4. **Spring-fall asymmetry:** Spring DTR (8.9°C) exceeds fall DTR (6.8°C) by 31%. This asymmetry likely results from ocean thermal inertia: the North Atlantic retains summer heat into fall, maintaining cloud cover and suppressing DTR, while spring experiences more variable air mass intrusions as the ocean warms.

3.4 DTR Time Series and Extremes

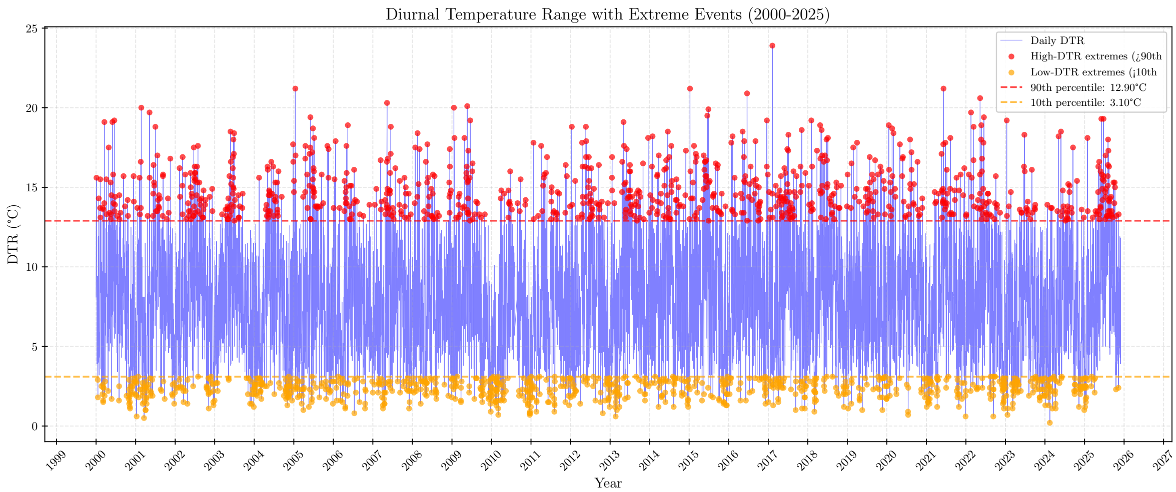

Figure 4: Daily DTR time series (2000–2025) with extreme events highlighted. Blue line shows daily DTR, red points indicate high-DTR extremes (>90th percentile = 11.04°C), orange points show low-DTR extremes (<10th percentile = 5.18°C). Horizontal dashed lines mark percentile thresholds. No temporal clustering of extremes is visually apparent.

Figure 4 displays the full daily DTR record with extreme events highlighted. High-DTR extremes and low-DTR extremes appear scattered throughout the time series without obvious temporal trends or clustering. The 90th and 10th percentile thresholds effectively capture the distribution tails while maintaining approximately 10% of observations in each extreme category as expected.

Visual inspection suggests high inter-annual variability in extreme event frequency, consistent with the high natural variability characteristic of maritime climates influenced by variable atmospheric circulation patterns.

3.5 Annual Extreme Event Frequency

Figure 5 quantifies annual extreme event frequencies. High-DTR extreme days range from approximately 15 to 52 per year, while low-DTR extremes range from 15 to 70 per year. This substantial variability (coefficient of variation ≈ 30–40%) reflects the influence of large-scale atmospheric modes such as the North Atlantic Oscillation (NAO) and blocking patterns that modulate regional weather patterns on inter-annual timescales.

No systematic increase or decrease in either extreme type is evident, though individual years (e.g., 2010, 2018) show notable spikes that may correspond to specific circulation anomalies.

3.6 Extreme Event Trend Analysis

Figure 6 presents trend tests for extreme event frequencies. Neither high-DTR nor low-DTR extremes exhibit statistically significant trends:

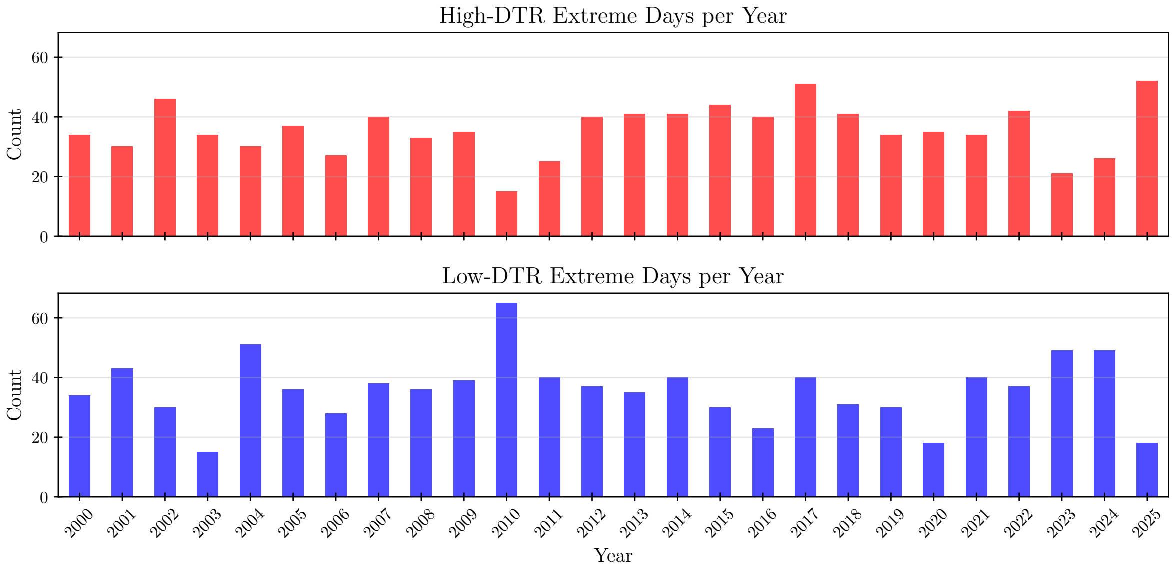

Figure 5: Annual counts of extreme DTR events (2000–2025). Top panel: high-DTR extremes (>90th percentile). Bottom panel: low-DTR extremes (<10th percentile). Both show substantial year-to-year variability with no apparent temporal trend. Some years experience 2–3× more extremes than others.

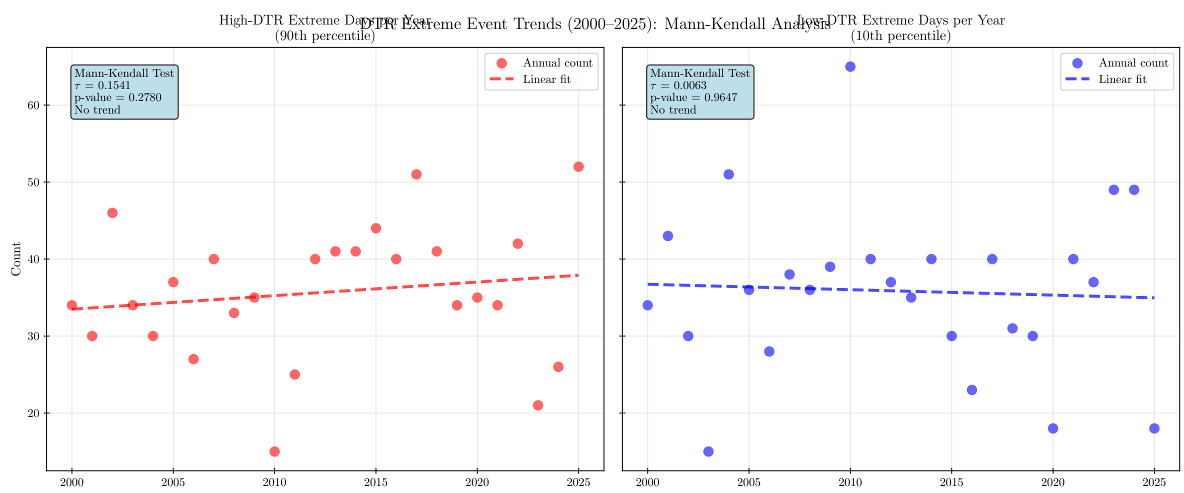

Figure 6: Mann-Kendall trend tests for extreme DTR events (2000–2025). Left panel: high-DTR extremes show $\tau = 0.154$ ($p = 0.447$), indicating no significant trend. Right panel: low-DTR extremes show $\tau = 0.006$ ($p = 0.811$), essentially flat. Blue/red annotation boxes indicate non-significant results. Linear regression fits (dashed lines) are shown for reference but do not imply causation.

- **High-DTR extremes:** $\tau = 0.154$, $p = 0.447$ (not significant, no trend)
- **Low-DTR extremes:** $\tau = 0.006$, $p = 0.811$ (not significant, no trend)

The near-zero Kendall's τ for low-DTR extremes indicates essentially flat behavior over time. The slightly positive τ for high-DTR extremes suggests a weak (non-significant) tendency toward more frequent clear-sky (trend) events, but with $p = 0.447$, this cannot be distinguished from random fluctuation.

Expected warming signature: Global warming theory predicts declining high-DTR extremes (fewer clear-sky days) and/or increasing low-DTR extremes (more cloudy/moist days). The absence of these coupled trends in St. John's suggests either:

1. The 26-year period is too short to detect trends amid high natural variability
2. Oceanic moderation dampens or delays the atmospheric response to warming
3. Regional circulation changes counteract global moisture increases

4 Discussion

4.1 Reconciling Warming with Stable DTR

The central paradox of this study is that St. John's exhibits significant warming (1.07°C) yet stable DTR. This contrasts with global patterns where warming typically produces declining DTR. Three physical mechanisms may explain this regional divergence:

4.1.1 Oceanic Thermal Inertia

The North Atlantic Ocean, particularly the cold Labrador Current, acts as a massive heat reservoir with high thermal capacity. This oceanic influence moderates both $T_{\max}$ and $T_{\min}$ increases, delays seasonal warming/cooling transitions and maintains persistent cloud cover through evaporative moisture supply.

If $T_{\max}$ and $T_{\min}$ are warming at similar rates due to oceanic buffering, DTR would remain stable despite overall temperature increases.

4.1.2 Atmospheric Circulation Changes

The North Atlantic Oscillation (NAO) and Arctic Oscillation (AO) modulate atmospheric circulation patterns over Atlantic Canada. Trends in these modes could alter the frequency of continental vs. maritime air mass intrusions, affecting DTR independently of mean temperature changes.

4.1.3 Short Time Series

Climate change signals typically require 30–50 years to emerge from natural variability. Our 26-year record may be insufficient to detect trends, particularly in a highly variable maritime setting where year-to-year fluctuations are large.

Table 1: Comparison of DTR trends across climate regimes

Location	Climate Type	DTR Trend	Reference
Global average	Mixed	Declining	Karl et al. (1993)
North America (continental)	Continental	Declining	Dai et al. (1999)
St. John's, NL	Maritime	Stable	This study

4.2 Comparison with Global Patterns

Table 1 contextualizes our findings. Continental regions show clear DTR declines, while St. John's remains stable. This supports the hypothesis that maritime climates respond differently to global forcing due to oceanic moderation. It is important to note that even though St. John's and the region may be stable, it most likely is only delaying the inevitable.

Table 1 contextualizes our findings. Continental regions show clear DTR declines, while St. John's remains stable. This supports the hypothesis that maritime climates respond differently to global forcing due to oceanic moderation. However, it is important to note that the observed stability may represent a delayed rather than absent response. Oceanic thermal inertia can buffer atmospheric changes on decadal timescales, potentially postponing but not preventing the eventual manifestation of global warming signals in DTR trends. Longer-term monitoring will be necessary to determine St. John's DTR stability.

4.3 Implications for Regional Climate Projections

While St. John's is warming, the stable diurnal temperature range means that day-to-day temperature swings remain consistent. For agriculture, this suggests that frost risk patterns and growing season conditions may not change as rapidly as in interior provinces like Saskatchewan or Manitoba, where larger DTR shifts are occurring. Energy demand patterns particularly heating needs in winter and cooling needs in summer depend on both overall temperature and the daily range between highs and lows. In St. John's, warming increases cooling demand but stable DTR means the pattern of when that demand occurs throughout the day remains predictable. Infrastructure planners also benefit from this stability in terms of the repeated expansion and contraction of materials (thermal stress cycles) on roads, bridges, and buildings follows familiar patterns, unlike continental regions where changing DTR alters maintenance schedules and design specifications.

From a climate science perspective, global climate models may overestimate DTR decline rates in maritime regions if they inadequately represent oceanic thermal inertia and coastal boundary layer processes. Downscaling efforts must account for ocean-atmosphere coupling to produce accurate local forecasts. Risk assessments for agriculture, infrastructure, and energy systems require region-specific analysis rather than applying global or continental patterns. The stable DTR despite warming also suggests that oceanic heat uptake is modulating atmospheric responses, with implications for understanding thermal stress on coastal and marine ecosystems, where air-sea temperature gradients drive critical processes like fog formation and upwelling.

Critically, if oceanic buffering is merely delaying and not preventing DTR decline, current stability provides a limited window for adaptation before rapid changes occur. The ocean's capacity to absorb heat and moderate climate change is finite. Continuous monitoring is essential to detect regime shifts that may emerge as ocean heat capacity becomes saturated, potentially triggering rapid changes in DTR patterns after decades of apparent stability.

4.4 Limitations

Several limitations should be acknowledged. First, using only two meteorological stations limits spatial representativeness; DTR patterns may vary across the broader St. John's region due to local topography, urban heat island effects, or proximity to coastline. Second, the 26-year time series may be insufficient for robust trend detection in a highly variable maritime climate where natural fluctuations can mask long-term signals. Third, our analysis assumes linear or monotonic trends; non-linear responses, abrupt regime shifts, or threshold effects would not be detected by Mann-Kendall tests or linear regression. Finally, we did not directly analyze the physical mechanisms driving DTR changes. Variables such as cloud cover fraction, atmospheric humidity or wind patterns were not incorporated, limiting our ability to attribute causality to observed patterns. Future work should extend the analysis to longer time series, incorporate additional meteorological variables, expand spatial coverage, and compare results with reanalysis datasets (e.g., ERA5) for validation.

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5 Conclusion

This study analyzed diurnal temperature range trends in St. John's, NL from 2000 to 2025 to test for climate change signals in a maritime setting. The analysis reveals a paradoxical pattern: while mean temperature increased significantly by 1.07°C over 26 years ($p < 0.001$) consistent with global warming trends, annual mean DTR shows no significant trend ($p = 0.229$), contrasting sharply with the global pattern of DTR decline. This stability persists despite clear warming, suggesting that maritime climates respond differently than continental regions.

Several key findings support this interpretation. First, the seasonal DTR amplitude of only 3.6°C indicates strong year-round oceanic moderation, far weaker than the 10+ °C amplitudes typical of continental sites. Second, neither high-DTR nor low-DTR extreme events show significant frequency changes over the study period, suggesting that the full temperature distribution, not just the mean remains stable in its variability structure. Third, spring DTR exceeds fall DTR by 31%, an asymmetry that reflects ocean thermal inertia: the North Atlantic retains summer heat into autumn, maintaining cloud cover and suppressing fall DTR, while spring experiences more variable air mass intrusions as the ocean warms.

The central conclusion is that regional climate responses to global forcings are modulated by oceanographic context. Maritime locations like St. John's may exhibit delayed or dampened responses compared to continental sites, highlighting the critical importance of region-specific analysis for climate adaptation planning. Coastal communities cannot simply extrapolate from continental or global trends when developing resilience strategies.

While global warming is evident in mean temperature trends, the expected DTR decline has not materialized in this 26-year maritime record. Whether this represents oceanic buffering that will eventually be overcome, insufficient time series length to detect an emerging trend, or genuine regional divergence from global patterns remains an open question. Answering it will require extending the analysis to longer time periods (50+ years), incorporating direct measurements of cloud cover and atmospheric circulation patterns, and comparing observational results with high-resolution climate model simulations that accurately represent ocean-atmosphere coupling in coastal regions.

Data and Code Availability

All data, analysis code, and figure generation scripts are available at:
`https://github.com/ieadoboe/cmsc6950-climate-daily`

References

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